



Blood Bank Management System

Frontend: HTML, CSS, Bootstrap, React · Backend: Java + Spring Boot ·
Database: MySQL

Extras: JWT auth · REST APIs · Google Maps · Cloud Storage



The Problem

Blood must be found fast – not searched manually.

Delay = Risk

Manual calls waste critical minutes

Fragmented Data

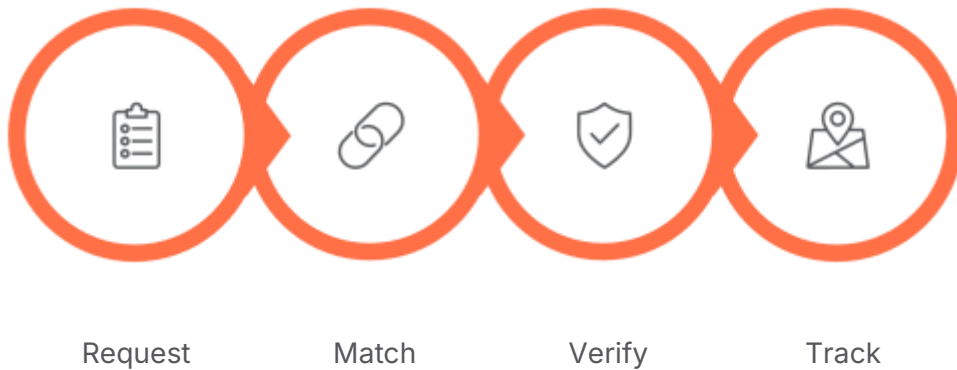
Donor records, inventory, and locations scattered

Need for automation

Fast matching + verified eligibility

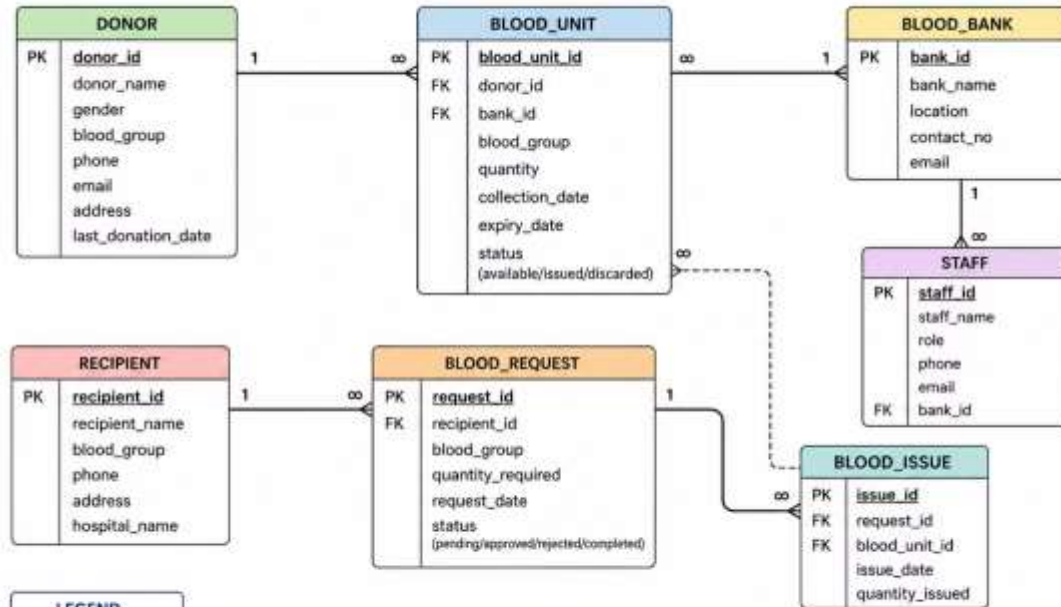


Core Workflow



Request blood → Smart match donors/stock → Medical verification →
Track bag to recipient

Blood Bank Management System – ER Diagram



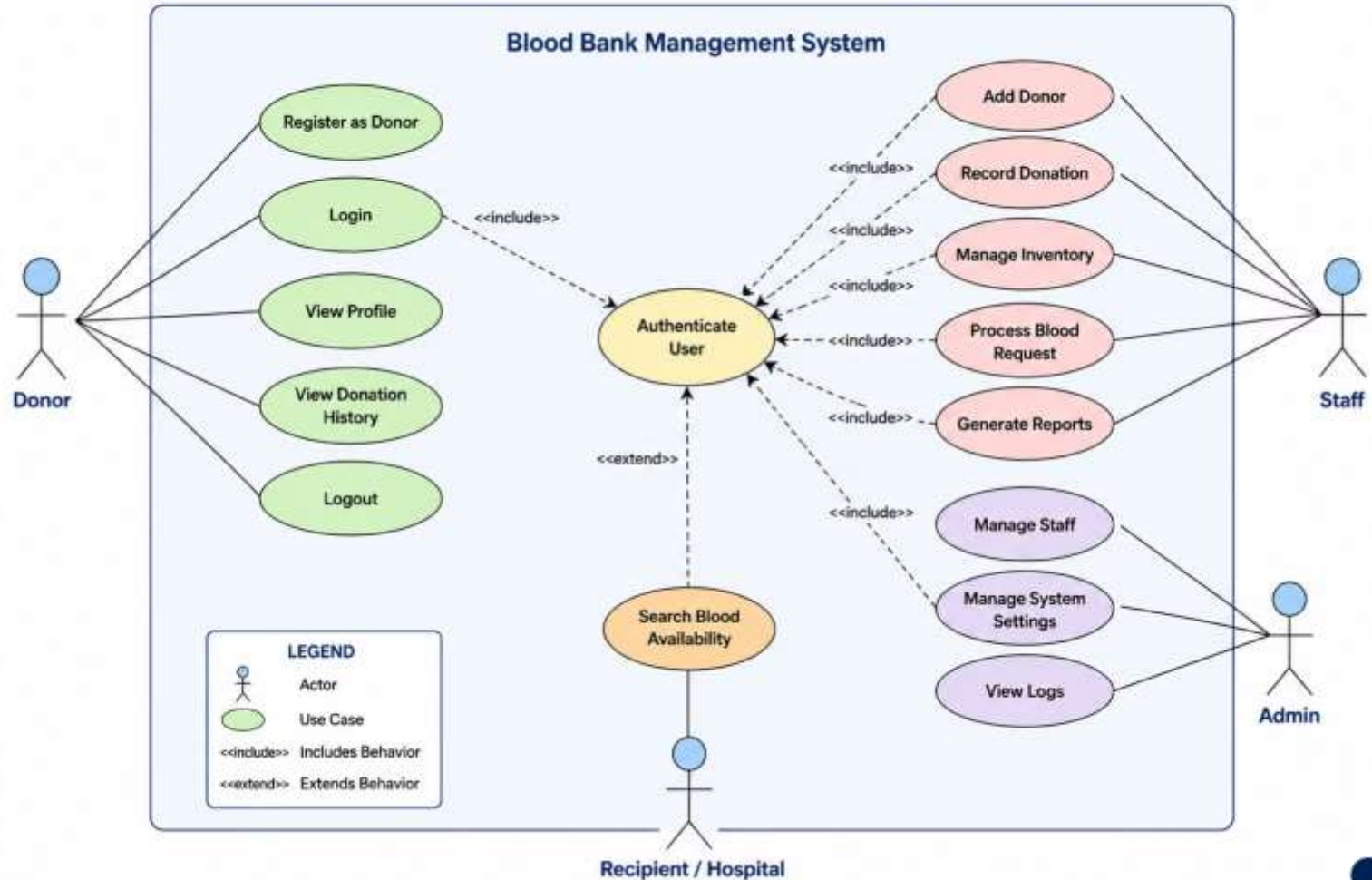
LEGEND

PK = Primary Key
 FK = Foreign Key
 1 = One
 ∞ = Many

ENTITY RELATIONSHIPS

- One **DONOR** can donate many **BLOOD_UNIT**.
- One **BLOOD_BANK** stores many **BLOOD_UNIT**.
- One **RECIPIENT** can make many **BLOOD_REQUEST**.
- One **BLOOD_REQUEST** can have one or more **BLOOD_ISSUE**.
- One **BLOOD_UNIT** can be used in many **BLOOD_ISSUE**.
- One **BLOOD_BANK** can have many **STAFF**.

Use Case Diagram – Blood Bank Management System



Location-based Matching

Google Maps API: lat/long proximity search · Nearby donor & blood center suggestions



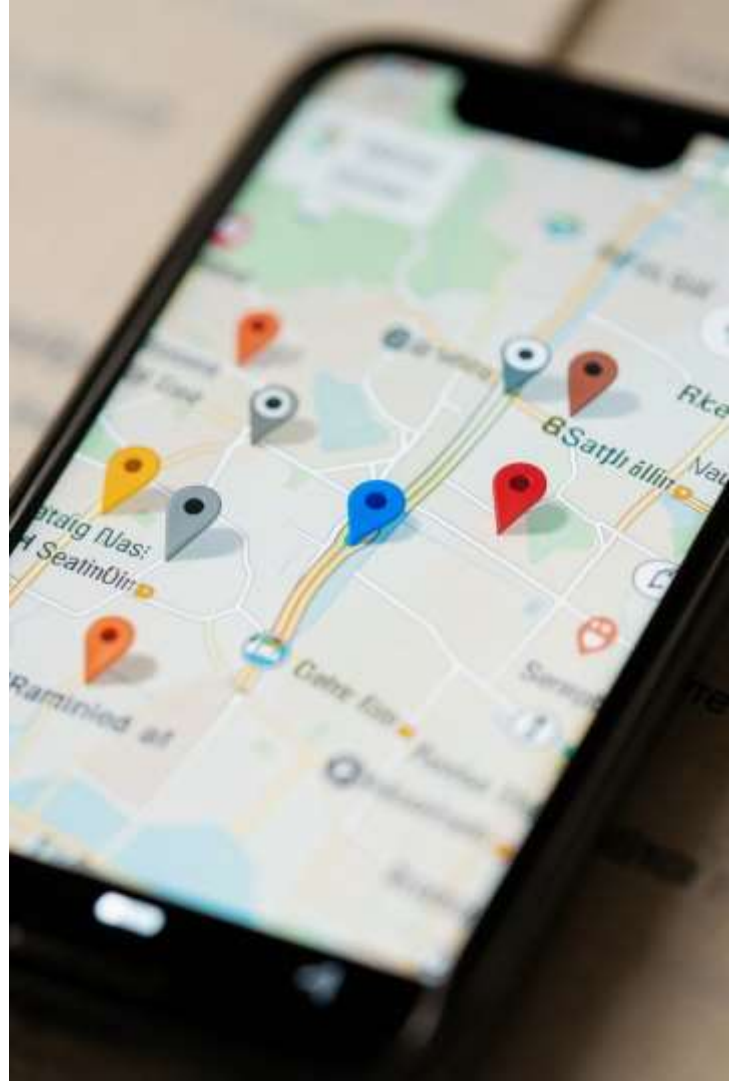
Emergency Mode

Closest matches prioritized



Route & ETA

Show fastest route for donor or courier



Security: JWT Authentication

Stateless tokens for React → Spring Boot APIs

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Role-based Access

Admin, Doctor, Donor scopes enforced

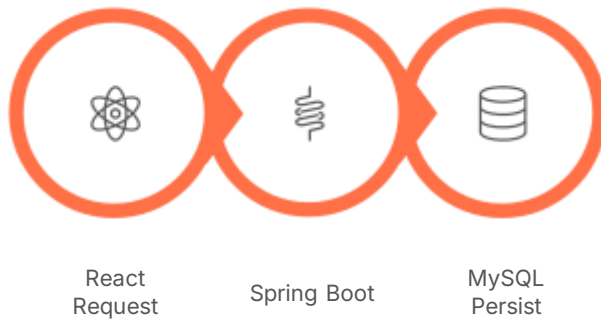
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Secure Endpoints

Token refresh, expiry, and revocation flows



REST APIs & Data



Modules: donors ·
blood · stock ·
requests · reports

- CRUD endpoints (Spring Boot)
- Validation & business rules (server side)
- MySQL for persistent, transactional data

Cloud Storage & Media

Store certificates, consent forms, donor photos

Keep React UI lightweight — serve assets from cloud



Secure Uploads

Signed URLs · Access controls



Fast Fetch

CDN-backed delivery

Conclusion — Time Saved = Lives Saved

Stack recap: HTML/CSS/Bootstrap/React · Java Spring Boot · MySQL · JWT · REST · Google Maps · Cloud Storage

Demo Flow



Urgent request → Map match → Secure update → Inventory confirm

Next Steps



Live demo, user testing, deployment

